

Foundations of Variational autoencoder

[Variational autoencoder]

$$\mathcal{L} = \mathbb{E}_{q_\phi}[\log p_\theta(x | z)] - D_{\text{KL}}(q_\phi(z | x) \| p(z))$$

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Overview of architecture and operation A variational autoencoder is a generative model with a prior and noise distribution respectively. Usually such models are trained using the expectation-maximization meta-algorithm (e.g. probabilistic PCA, (spike & slab) sparse coding). Such a scheme optimizes a lower bound of the data likelihood, which is usually computationally intractable, and in doing so requires the discovery of q-distributions, or variational posteriors. These q-distributions are normally parameterized for each individual data point in a separate optimization process. However, variational autoencoders use a neural network as an amortized approach to jointly optimize across data points. In that way, the same parameters are reused for multiple data points, which can result in massive memory savings. The first neural network takes as input the data points themselves, and outputs parameters for the variational distribution. As it maps from a known input space to the low-dimensional latent space, it is called the encoder. The decoder is the second neural network of this model. It is a function that maps from the latent space to the input space, e.g. as the means of the noise distribution. It is possible to use another neural network that maps to the variance, however this can be omitted for simplicity. In such a case, the variance can be optimized with gradient descent. To optimize this model, one needs to know two terms: the "reconstruction error", and the Kullback–Leibler divergence (KL-D). Both terms are derived from the free energy expression of the probabilistic model, and therefore differ depending on the noise distribution and the assumed prior of the data, here referred to as p-distribution. For example, a standard VAE task such as IMAGENET is typically assumed to have a gaussianly distributed noise; however, tasks such as binarized MNIST require a Bernoulli noise. The KL-D from the free energy expression maximizes the probability mass of the q-distribution that overlaps with the p-distribution, which unfortunately can result in mode-seeking behaviour. The "reconstruction" term is the remainder of the free energy expression, and requires a sampling approximation to compute its expectation value. More recent approaches replace Kullback–Leibler divergence (KL-D) with various statistical distances, see "Statistical distance VAE variants" below.

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the sliced Wasserstein distance used by S Kolouri, et al. in their VAE the energy distance implemented in the Radon Sobolev Variational Auto-Encoder the Maximum Mean Discrepancy distance used in the MMD-VAE the Wasserstein distance used in the WAEs kernel-based distances used in the Kernelized Variational Autoencoder (K-VAE)

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Statistical distance VAE variants After the initial work of Diederik P. Kingma and Max Welling, several procedures were proposed to formulate in a more abstract way the operation of the VAE. In these approaches the loss function is composed of two parts :

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$$\mathcal{L}_{\theta, \phi} = \mathbb{E}_{x \sim P_{\text{real}}} [\|x - D_{\theta}(\mathbb{E}_{\phi}(x))\|_2^2] + d(\mu(D_z), \mathbb{E}_{\phi} P_{\text{real}})^2 \quad \{\displaystyle \mathcal{L}_{\{\theta\}, \{\phi\}} = \mathbb{E}_{x \sim \mathbb{P}} \|\mathbb{E}_{\phi}(x) - x\|_2^2 + d(\mu(D_z), \mathbb{E}_{\phi} P)^2\}$$

$$\int \mathbb{E}_{p_{\theta}(x|z)} \left[\frac{1}{2} \left(\frac{\partial \log p_{\theta}(x|z)}{\partial z} \right)^2 \right] dz$$

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In the vanilla variational autoencoder, z is usually taken to be a finite-dimensional vector of real numbers, and $p_{\theta}(x|z)$ to be a Gaussian distribution. Then $p_{\theta}(x)$ is a mixture of Gaussian distributions. It is now possible to define the set of the relationships between the input data and its latent representation as

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This can be reparametrized by letting $\epsilon \sim N(0, I)$ be a "standard random number generator", and construct z as $z = \mu_{\phi}(x) + L_{\phi}(x)\epsilon$. Here, $L_{\phi}(x)$ is obtained by the Cholesky decomposition: $\Sigma_{\phi}(x) = L_{\phi}(x)L_{\phi}(x)^T$. Then we have $\nabla_{\phi} \mathbb{E}_{z \sim q_{\phi}(\cdot|x)} [\ln p_{\theta}(x, z) q_{\phi}(z|x)] = \mathbb{E}_{z \sim q_{\phi}(\cdot|x)} [\nabla_{\phi} \ln p_{\theta}(x, \mu_{\phi}(x) + L_{\phi}(x)\epsilon) q_{\phi}(\mu_{\phi}(x) + L_{\phi}(x)\epsilon|x)]$. Then we have $\nabla_{\phi} \mathbb{E}_{z \sim q_{\phi}(\cdot|x)} [\ln \frac{p_{\theta}(x, z) q_{\phi}(z|x)}{p_{\theta}(x, \mu_{\phi}(x) + L_{\phi}(x)\epsilon) q_{\phi}(\mu_{\phi}(x) + L_{\phi}(x)\epsilon|x)}] = \mathbb{E}_{z \sim q_{\phi}(\cdot|x)} [\nabla_{\phi} \ln \frac{p_{\theta}(x, \mu_{\phi}(x) + L_{\phi}(x)\epsilon) q_{\phi}(\mu_{\phi}(x) + L_{\phi}(x)\epsilon|x)}{p_{\theta}(x, z) q_{\phi}(z|x)}]$ and so we obtained an unbiased estimator of the gradient, allowing stochastic gradient descent. Since we reparametrized z , we need to find $q_{\phi}(z|x)$. Let $q_0(\epsilon)$ be the probability density function for ϵ , then $\ln q_{\phi}(z|x) = \ln q_0(\epsilon) - \ln |\det(\partial z / \partial \epsilon)|$ where $\partial z / \partial \epsilon$ is the Jacobian matrix of z with respect to ϵ . Since $z = \mu_{\phi}(x) + L_{\phi}(x)\epsilon$, this is $\ln q_{\phi}(z|x) = -\frac{1}{2} \epsilon^T \Sigma_{\phi}^{-1}(x) \epsilon - \ln |\det L_{\phi}(x)| - \frac{n}{2} \ln (2\pi)$

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where $p_{\theta}(x, z)$ represents the joint distribution under p_{θ} of the observable data x and its latent representation or encoding z . According to the chain rule, the equation can be rewritten as